

GUARDIAN SYSTEM

Doorbell Module

3D-Printed, RFID-Based Doorbell Hardware Documentation

This is an in-depth documentation of the Guardian System doorbell module.



Completed doorbell module with the top cover installed.

1. Module Overview

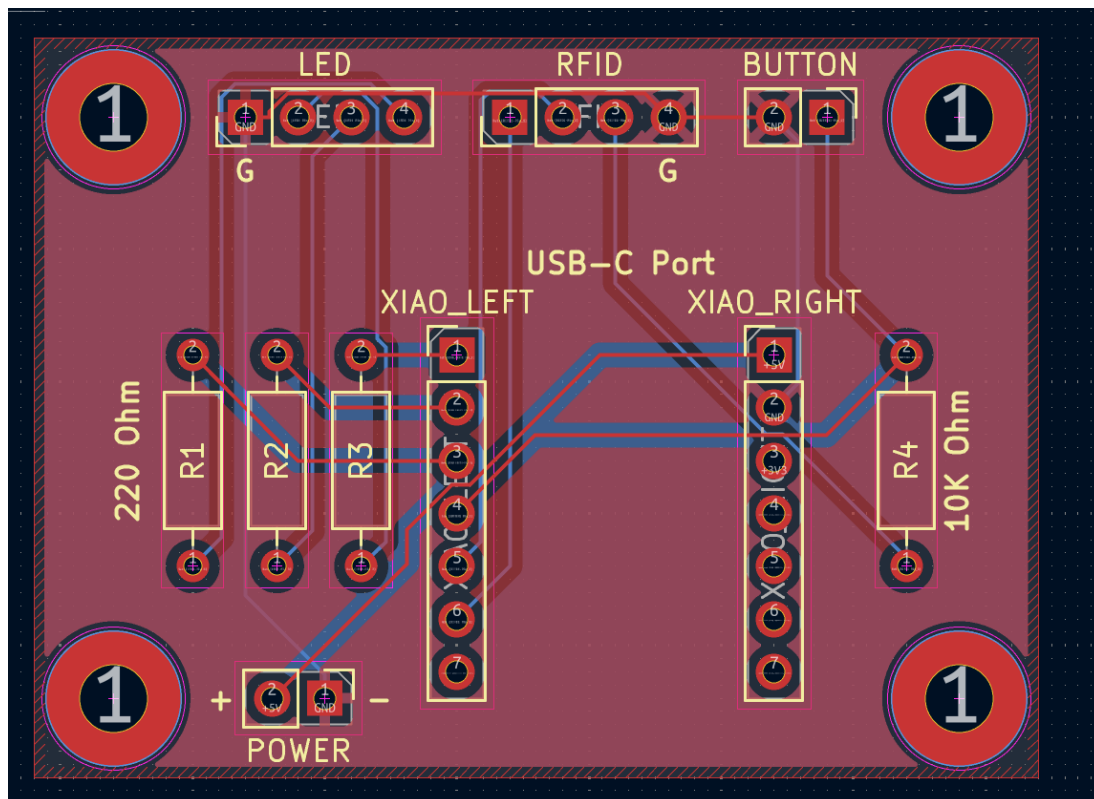
The doorbell module combines 3D-printed parts and a custom PCB. The PCB supports an ESP32, an RFID module, a push button, and an optional RGB LED. The module is intended to work as part of the Guardian System and can be integrated with Home Assistant.

3D-printing requirement: Print the provided parts exactly as supplied. No modifications are needed. The support structures are already prepared, and the print settings are optimized for PETG prints.

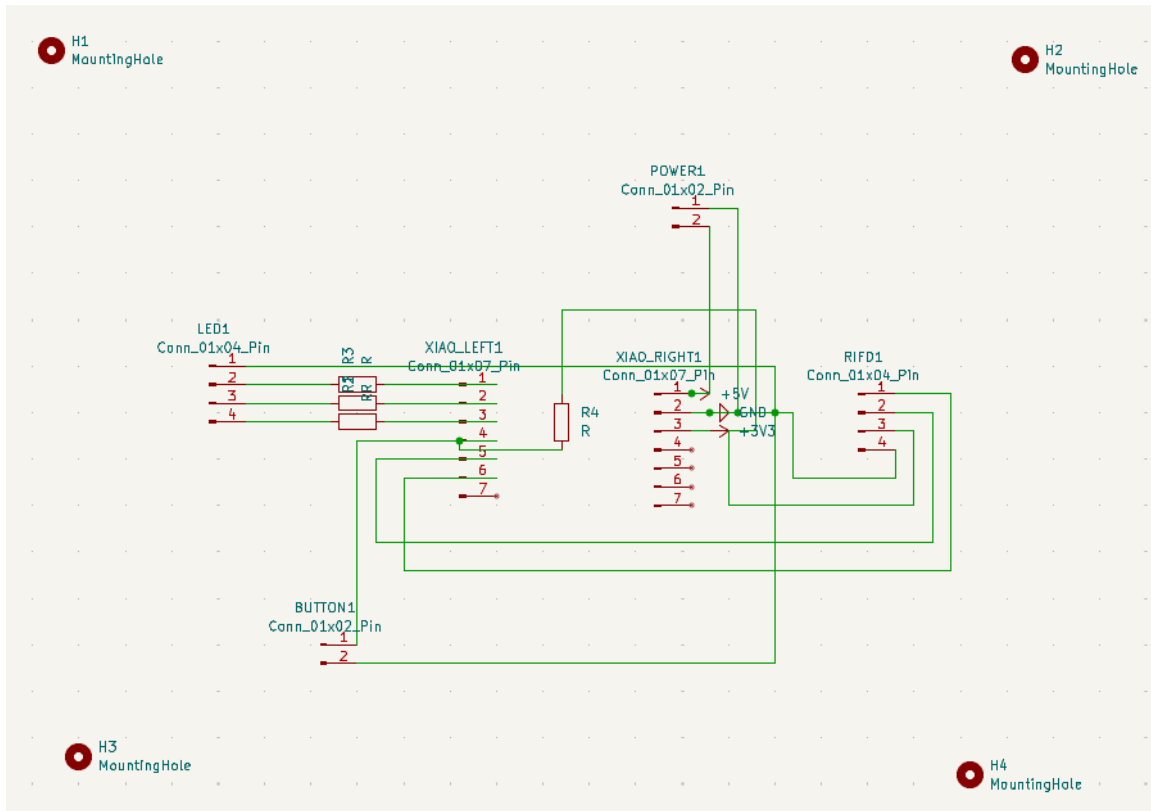
Document workflow

1. Prepare the hardware and printed parts.
2. Assemble and solder the doorbell PCB.
3. Install the PCB and modules into the printed enclosure.
4. Mount the assembled doorbell into the wall opening.

2. PCB Reference



PCB layout showing the LED, RFID, button, ESP32, resistor, and power positions.



Doorbell PCB schematic.

3. Hardware Needed

Quantity	Component	Purpose / note
2	1x7 female pin headers	Connector option; see the connector note below.
2	1x4 female pin headers	Connector option; see the connector note below.
2	1x2 female pin headers	Connector option; see the connector note below.
3	220 Ohm THT resistors	For the LED.
1	10K Ohm resistor	For the push button, as listed in the hardware list.
15+	M3 threaded inserts	Used throughout the printed enclosure and mounting parts.
1	RFID module	RFID reader module.
1	Push button	Momentary 12 mm red push button.
1	ESP32	Seeed Studio XIAO ESP32-C6.
1 optional	5 mm RGB LED, common anode	Used only for troubleshooting and

		showing the doorbell connection status.
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Connector note: Any connector that fits the 2.54 mm hole spacing can be used. Female headers can help avoid connector-orientation mistakes. The ESP32 may also be soldered directly to the PCB using male pin headers.

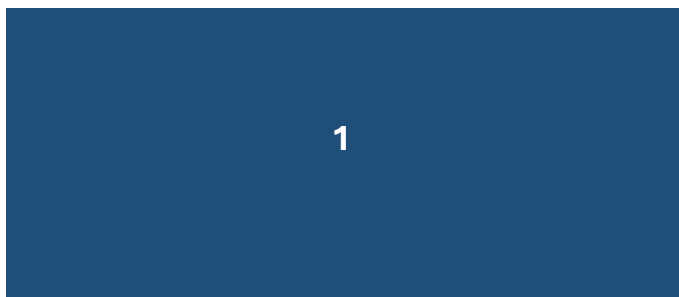
Resistor-value check: The hardware list specifies one 10K Ohm resistor for the push button, while the PCB soldering instructions later refer to a single 1K resistor on the right side of the ESP32. Both statements are retained here; check the PCB marking before soldering.

Components Used in the Original Build

The following links point to the local Greek technology shop used for the original build:

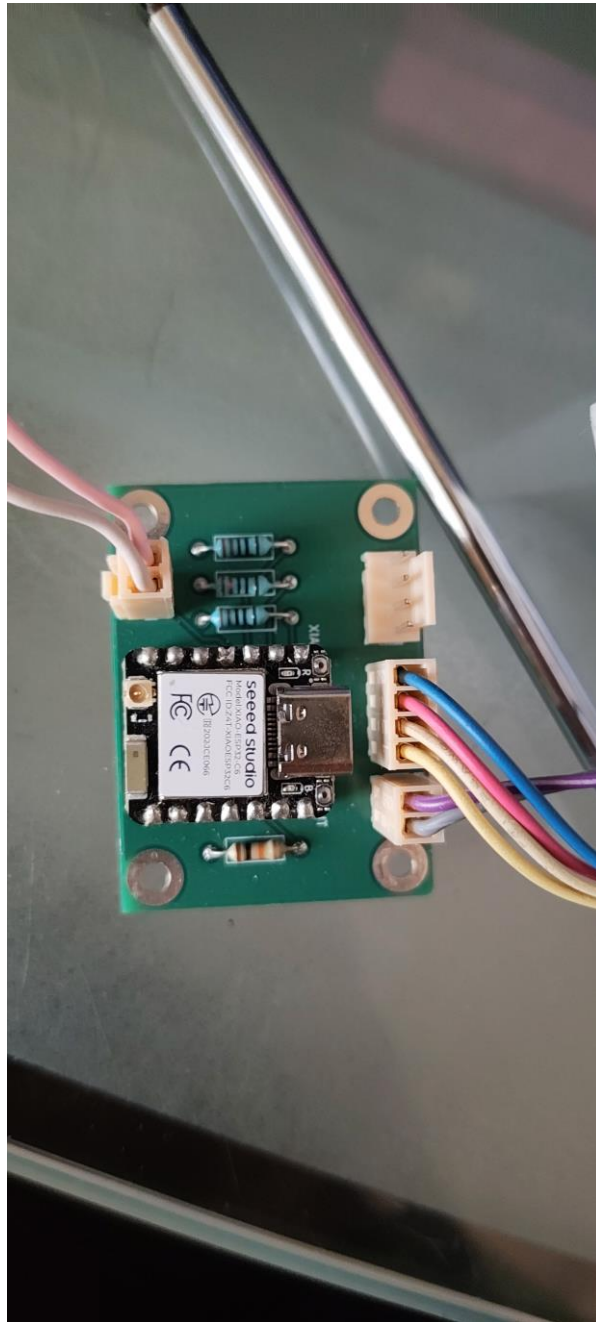
- M3 threaded inserts: <https://grobotronics.com/threaded-insert-m3-5-7x4-6mm.html>
- RFID module: <https://grobotronics.com/acebott-easy-plug-rfid-rc522-module.html>
- Push button: <https://grobotronics.com/push-button-momentary-12mm-red.html>
- ESP32: <https://grobotronics.com/seeed-studio-xiao-esp32-c6.html>
- RGB LED (optional): <https://grobotronics.com/led-diffused-5mm-rgb-common-anode.html>

4. PCB Assembly



Install the ESP32

Solder the ESP32 directly to the PCB, or install female headers on the PCB and male headers on the ESP32. Make sure the USB-C port faces the RFID connector. The PCB includes markings to show the correct orientation.



Assembled PCB showing the ESP32 orientation, resistors, connectors, and power input.

2

Solder the resistors

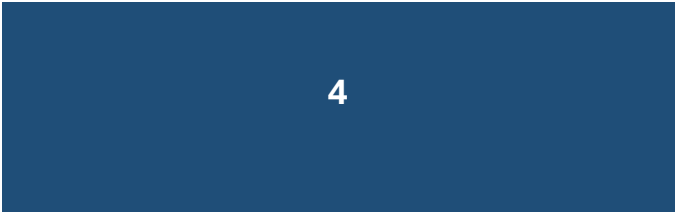
Solder the three 220 Ohm resistors to the left of the ESP32. Solder the single 1K resistor to the right of the ESP32, as stated in the original assembly instructions.



3

Install the remaining connectors

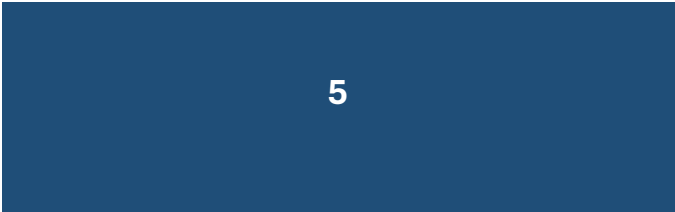
Solder the other connectors in any order, while carefully checking their orientation. Follow each cable to its end. The wire from the ESP32 GND pin must connect to the PCB pin marked G. Apply the same check to the LED connection.



4

Connect the push button

The push-button connector orientation is not important; either orientation produces the same result.



5

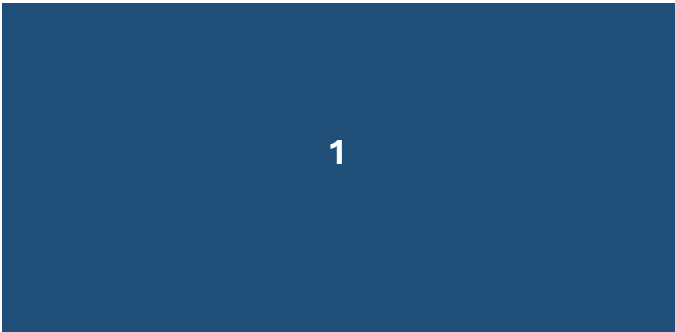
Connect power

Power polarity is important. Connect the 5V cable from a regular 5V power supply to the PCB pin marked 5V. Connect the other power pin to ground.

PCB assembly complete: After the ESP32, resistors, connectors, push button, LED connection, and power input are installed as described above, the doorbell PCB assembly is complete.

5. Final Mechanical Assembly

Wall-opening requirement: The hole used for the doorbell must be at least 54 mm in diameter and at most 45 mm deep. Otherwise, mounting issues may occur.



1

Install the first threaded inserts

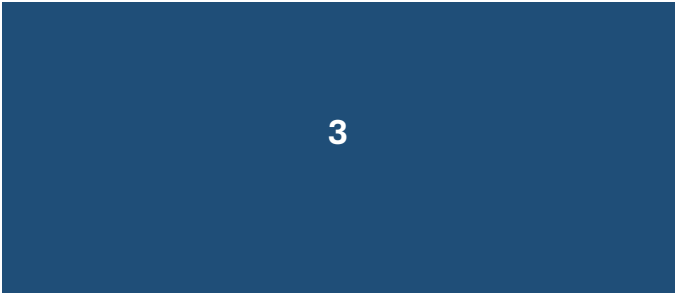
After printing is complete, place M3 threaded inserts into the four holes in the cone-like area of the main doorbell part. Position each insert in its hole. Instead of inserting the soldering iron through the top holes, touch the insert with the iron tip from the large rear opening. Heat it slowly until it moves into place.



2

Prepare the cables before seating the PCB

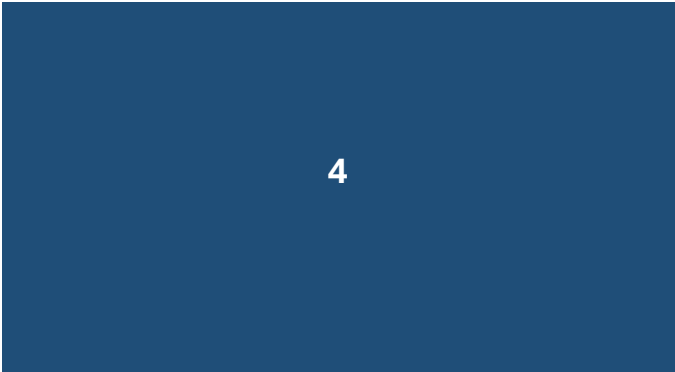
Connect the cables before placing the PCB into position. This is especially important for the power cable, because connecting it later may require moving the board to the empty side and then moving it back.



3

Prepare the push button early

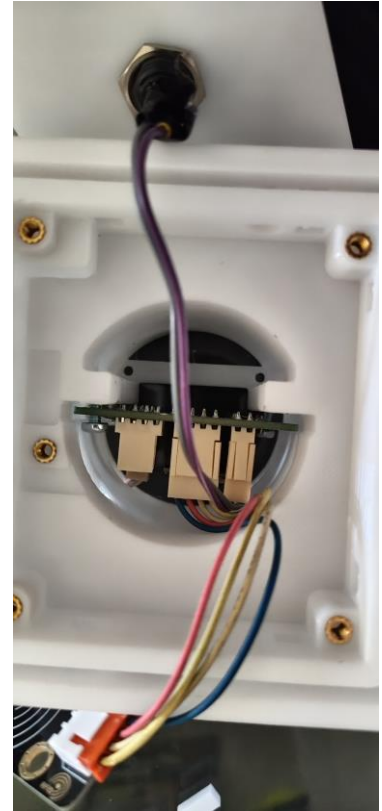
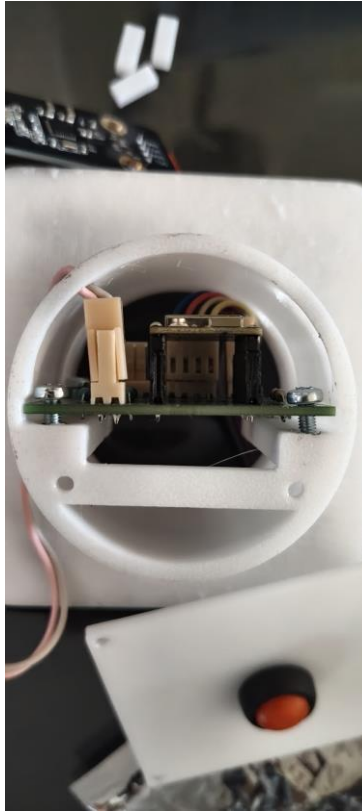
Before connecting the push-button cables, screw the push button to the rectangular 3D-printed part. The button and its cable must be completely free during this process, so this step should be completed early.



4

Place and loosely secure the PCB

Place the PCB over the threaded-insert holes with the power pins toward the edge, allowing the cable to be routed through the screw openings. Start fastening the PCB, but do not tighten it fully because the pins extend through the opposite side. It may be difficult to use every screw hole; at least two screw holes are needed.



PCB and RFID module positioning inside the printed enclosure.

5

Route the cables and install the RFID module

Route the cables toward the front. Install M3 threaded inserts in the available holes. Screw the RFID module to the single insert positioned farther inside, as shown in the assembly images.

6

Install the optional RGB LED

The RGB LED is optional and is used only for troubleshooting and showing the doorbell connection status. Place it in the circular hole at the bottom side of the doorbell, based on the perspective shown in the images. Insert it only lightly; do not push it all the way in, because it may interfere with the top cover.

7

Install the push-button plate

Position the rectangular push-button part over the threaded inserts and screw it into place.



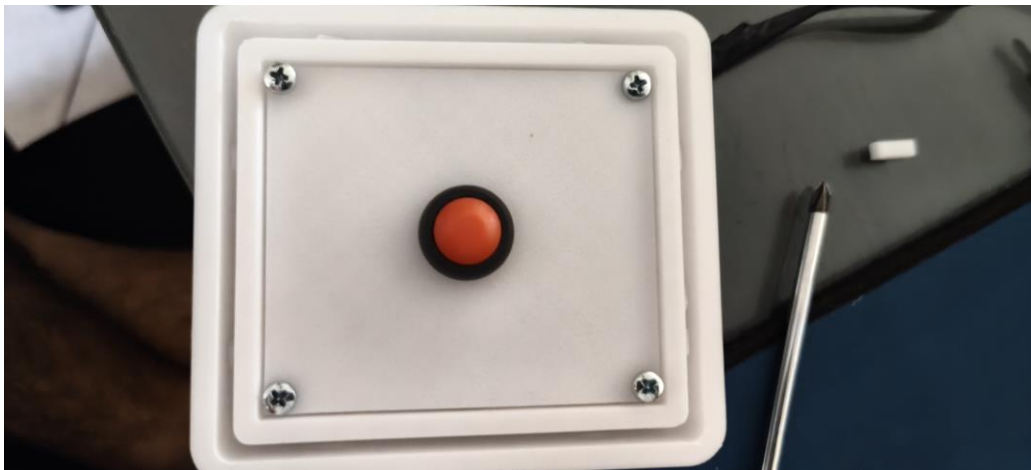
Side view of the push-button plate during installation.

8

Align the top cover

Position the top cover so that its pill-shaped opening aligns with the circular opening on the other part. Place the small rectangular pieces into the openings on both sides of the doorbell body. To remove the cover later, push these pieces inward so the cover can be taken off.

Do not install the top cover yet: Complete the wall mounting first. Install the top cover only after the doorbell module is fully mounted in the wall opening.



Front view of the push-button plate and cover area.

6. Wall Mounting

1

Prepare the rear mounting part

Install M3 threaded inserts into the bottom holes of the final unused printed part. The middle holes are used for wall mounting.

2

Attach the rear part to the wall opening

Place the rear part behind the opening with the threaded-insert holes on the underside. Screw it into place using the mounting method suitable for the installation.

3

Secure the doorbell module

With the top cover removed, place the doorbell module into the opening. Use a long screwdriver to install M3 screws through the rear holes of the module.



Rear mounting part with two M3 threaded inserts installed in the lower holes.

4

Install the top cover

After the module is fully mounted, install the top cover using the alignment and side-retainer method described in the previous section.



Final assembled doorbell module.